Daily Calculus Drill 📐

Technical Architecture, AI Engine & Developer Specification Sheet

React Native 0.88.1 | Expo SDK 58 | TypeScript 6.0 | SQLite (WAL Mode) | On-Device Neuro-Symbolic AI | mathjs Symbolic CAS

1. Abstract & High-Level Overview

Daily Calculus Drill is a cross-platform, 100% offline-capable mobile habit tracker and adaptive calculus puzzle application. Every morning (default `07:00 AM`), the system triggers a local push notification inviting the user to solve a daily derivative or indefinite integral.

Rather than using static question banks or requiring paid cloud LLM APIs, the app embeds a **Tiny Local Neuro-Symbolic AI Engine** paired with the **mathjs Computer Algebra System (CAS)**. It dynamically synthesizes calculus problems according to the user's selected experience tier (*Beginner, Intermediate, Advanced, Master*), tracks per-topic mastery using **Bayesian Knowledge Tracing (BKT)** and an **Elo Skill Rating**, and verifies algebraic equivalence symbolically and numerically.

2. Layered System Architecture

1. Presentation Layer (UI & Themes)

Built with React Native functional components, `DARK_PALETTE` (default) & `LIGHT_PALETTE` theme tokens, animated shake feedback, KaTeX WebView typography (with instant Unicode fallback), and a custom math symbol toolbar.

2. State & Orchestration Layer

`DrillContext.tsx` coordinates startup database synchronization, midnight date rollover, live solve timer, optimistic 3-attempt tracking (`3/3 → 2/3 → 1/3 → FAILED`), onboarding modal state, and theme switching.

3. On-Device Intelligence Layer

`localMathAIModel.ts` & `mathVerifier.ts` implement Contextual Multi-Armed Bandit topic selection, symbolic AST synthesis via `mathjs.derivative()` , LaTeX sanitization, and multi-point numerical equivalence checking.

4. Persistence & Notification Layer

`db.ts` manages an asynchronous SQLite database (`daily_calculus_drill.db`) with automatic versioned schema migrations (v5). `notificationService.ts` manages daily local system-tray scheduling.

3. Technology Stack Specification

Component	Package	Version	Architectural Responsibility
App Runtime	`expo` / `react-native`	`~58.0.23` / `0.88.1`	Cross-platform native execution for Android, iOS, and Web.

Component	Package	Version	Architectural Responsibility
Database	`expo-sqlite`	`~58.0.15`	Local SQLite storage in WAL mode for progress, AI mastery, and history.
Symbolic CAS	`mathjs`	`^14.0.0`	Symbolic differentiation (`derivative`), simplification (`simplify`), and evaluation.
Math Rendering	`react-native-webview`	`14.0.1`	Renders LaTeX equations via KaTeX with zero-flicker Unicode fallback.
Notifications	`expo-notifications`	`~58.1.8`	Daily recurring 07:00 AM local notification & deep-link response listener.
Navigation	`@react-navigation/*`	`^7.0.14`	Bottom Tab navigation (`DailyDrill` , `History` , `Settings`) & header actions.

4. Codebase Directory Structure

```
c:\Users\R E V O\Desktop\testAnti\
├── App.tsx                                  # Root Navigation, "?" Help Header Button & Theme Wrapper
├── app.json / package.json / tsconfig.json # Expo SDK 58 & TypeScript Configuration
└── src/
    ├── types/index.ts                       # Data Models (UserProgress, DrillHistory, CalculusProblem,
ThemeMode)
    ├── theme/colors.ts                      # DARK_PALETTE (default) & LIGHT_PALETTE definitions
    ├── ai/localMathAIModel.ts               # Local Neuro-Symbolic AI Generator, Elo Rating & BKT
Mastery Engine
    ├── utils/
    │   ├── problemGenerator.ts              # Generator bridge connecting user state to the Local AI
Model
    │   └── mathVerifier.ts                  # LaTeX sanitizer, +C checker, symbolic & numerical
equivalence engine
    ├── database/db.ts                       # SQLite initialization, v5 schema migration, rollover &
CRUD helpers
    ├── services/notificationService.ts      # Daily 07:00 AM notification scheduler & test trigger
    ├── context/DrillContext.tsx             # Global state provider, 3-attempt lifecycle & theme state
    ├── components/
    │   ├── MathRenderer.tsx                 # KaTeX WebView equation display + offline Unicode fallback
    │   ├── MathKeyboardToolbar.tsx          # Quick-insert calculus symbol chips (+ C, ^, √, e^x, ln,
trig)
    │   └── HowToPlayModal.tsx               # First-time onboarding popup with "Do not show again" &
close button
    └── screens/
        ├── DailyDrillScreen.tsx             # Home challenge UI, difficulty selector, shake anim &
Level-Up modal
        ├── HistoryScreen.tsx                # 14-day calendar activity strip, stats & chronological
attempt log
        └── SettingsScreen.tsx               # Theme radio selector, AI mastery bars, time picker,
backup & reset
```

5. SQLite Database Schema (`src/database/db.ts`)

The SQLite database runs with `PRAGMA journal_mode = WAL;` and uses an automatic migration tracker (`SCHEMA_VERSION = 5`) so new columns are added non-destructively via `ALTER TABLE` .

5.1. `user_progress` Table (Single-Row Global State, `CHECK (id = 1)`)

Column Name	SQLite Type	Default	Purpose
`id`	`INTEGER PRIMARY KEY`	`1`	Single-row constraint (`CHECK (id = 1)`).
`current_level`	`INTEGER NOT NULL`	`1`	User's current stage; increments `+1` on every solved problem.
`streak`	`INTEGER NOT NULL`	`0`	Consecutive days solved; resets to `0` on failure or missed day.
`last_attempt_date`	`TEXT`	`NULL`	Local date string (`YYYY-MM-DD`) of most recent attempt.
`notification_hour`	`INTEGER NOT NULL`	`7`	Daily reminder hour (24-hour format, default `07:00 AM`).
`notification_minute`	`INTEGER NOT NULL`	`0`	Daily reminder minute (`0..59`).
`selected_difficulty`	`TEXT NOT NULL`	`'BEGINNER'`	Experience tier: `'BEGINNER' | 'INTERMEDIATE' | 'ADVANCED' | 'MASTER'` .
`skill_rating`	`INTEGER NOT NULL`	`800`	Adaptive Elo skill rating (`400..2600`).
`topic_mastery_json`	`TEXT NOT NULL`	`JSON`	Serialized mastery weights (`0.10..0.98`) across 6 calculus topics.
`hide_onboarding_modal`	`INTEGER NOT NULL`	`0`	`0` = show How-To-Play popup on launch; `1` = do not show again.
`theme_mode`	`TEXT NOT NULL`	`'DARK'`	Active theme mode (`'DARK'` default or `'LIGHT'`).

5.2. `drill_history` Table (Chronological Daily Records)

Column Name	SQLite Type	Description
`id`	`INTEGER PRIMARY KEY AUTOINCREMENT`	Unique record ID.
`date`	`TEXT NOT NULL UNIQUE`	Local date key (`YYYY-MM-DD`).
`level`	`INTEGER NOT NULL`	Level number attempted.
`problem_latex`	`TEXT NOT NULL`	LaTeX equation string (e.g., `\int 6x^2 \, dx`).

Column Name	SQLite Type	Description
`correct_answer`	`TEXT NOT NULL`	Canonical `mathjs` expression (e.g., `2*x^3 + C`).
`user_answer`	`TEXT NOT NULL`	Latest expression submitted by the user.
`status`	`TEXT NOT NULL`	`'PENDING' | 'SOLVED' | 'FAILED' | 'SKIPPED'`.
`attempts_used`	`INTEGER NOT NULL`	Attempts consumed (`0`, `1`, `2`, or `3`).
`duration_seconds`	`INTEGER`	Elapsed solve duration in seconds.
`category` / `hint`	`TEXT`	Topic category display label and step-by-step hint text.
`ai_reasoning`	`TEXT`	Explanation generated by the Local AI Coach for this problem.
`difficulty` / `topic_key`	`TEXT`	Experience tier and internal topic key used for BKT mastery updates.

6. Local Neuro-Symbolic AI Engine (`src/ai/localMathAIModel.ts`)

The on-device AI engine personalizes problem generation without external API calls using three mathematical mechanisms:

1. **Experience Tier Profiles (`DIFFICULTY_PROFILES`):**
   - **Beginner (Base Elo 800):** Power rule derivatives & basic polynomial integrals.
   - **Intermediate (Base Elo 1150):** Trigonometric (`sin, cos`), exponential (`e^(kx)`), logarithmic (`ln(x)`), and multi-term polynomials.
   - **Advanced (Base Elo 1500):** Product rule, quotient rule, and u-substitution integrals.
   - **Master (Base Elo 1850):** Integration by parts (`∫ kx e^x dx`, `∫ k ln(x) dx`) and composite chain rule.
2. **Contextual Bandit Topic Selection (`selectTargetTopicWithAI`):** Scores each allowed topic t to prioritize the user's weakest topics while penalizing immediate repetition:

   ```
   Utility(t) = 0.75 × (1 - Mastery[t]) - 0.15 × IsPreviousTopic(t) +
   ExplorationNoise(0..0.25)
   ```

3. **Bayesian Mastery & Elo Rating Update (`updateSkillModelAfterAttempt`):** Scales coefficients with `skill_rating`, computes symbolic derivatives via `mathjs.simplify(mathjs.derivative(expr, 'x'))`, and updates Elo after each drill (+10 to +34 on solve depending on attempt count and speed; -22 on fail).

7. Algebraic Verification & Core User Workflows

- **Two-Stage Equivalence Verification (`src/utils/mathVerifier.ts`):**
  1. *Sanitization:* Converts LaTeX/shorthand (`\sqrt{u} → sqrt(u)`, `ln(u) → log(u)`, `c → C`, `\pi → pi`) into valid `mathjs` syntax and detects missing `+ C` on indefinite integrals.
  2. *Stage 1 (Symbolic):* Evaluates `mathjs.simplify(`(${user}) - (${expected})`).toString() === '0'`.

3. *Stage 2 (Multi-Point Numerical):* Compiles both expressions and evaluates across 5 test points (`x ∈ {0.41, 0.83, 1.27, 1.79, 2.31}, C = 4.25`) within relative tolerance `10^-6` to verify trigonometric and factored identities.

- **3-Attempt Progression Lifecycle:** Every puzzle starts at `3 / 3` attempts. Each non-empty wrong submission triggers an error shake animation and decrements remaining attempts (`3/3 → 2/3 → 1/3 → FAILED`). Solving within 3 attempts increments `current_level + 1` and `streak + 1` .
- **Onboarding Guide & Theme Switcher:** First-time users receive a *How to Play* popup with a "*Do not show again on startup*" checkbox and a persistent `?` button in the top-right header. Settings provides radio buttons to switch between **Dark Theme (Default)** and **Light Theme**.